

SINGLE AND MULTIPLE DISCRIMINATION REGIONS IN
MULTIPLE LINEAR REGRESSION

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1. Introduction

In this paper the problem is studied of discrimination or inverse prediction in multiple linear regression models. A generalization of the classical problem of discrimination in single regression, namely a problem of simultaneous inference in the reverse direction (of prediction) was solved and described by Miller (1966). The model is simple linear regression (one independent variable) and observations $Y_1^0, \dots, Y_k^0$ are given for which it was required to estimate the corresponding unknown values of the independent variable, $X_1^0, \dots, X_k^0$ and to find a set of confidence regions for these. The case $k=1$ was the original problem solved in this respect and has been described, for example by Brownlee (1960). This same problem was treated more explicitly by Laubscher (1962) for the case of parabolic regression. A further related problem which is relevant has been discussed by Lieberman, Miller and Hamilton (1967). We do not intend to discuss this problem in the present context.

It is our purpose rather to treat here the case of inverse inference for the multiple linear regression model. We shall mainly concern ourselves with the non-simultaneous problem, i.e. that in which we are required to discriminate for only one value of the dependent variable. Also, to obtain an insight into some of the geometry of the problem we shall discuss in more detail the case where there are only 2 independent variables.

2. Discrimination with two independent variables

2.1 Analysis of the model

The model we study is the usual multiple regression model specialized to the case of 2 independent variables X_1 and X_2 . The data available consist of N triplets $(Y_i, X_{1i}, X_{2i}), i=1, \dots, N$ where Y_i is the observed response for given X_{1i} and X_{2i} . The form in which we prefer to consider this regression model is

$$\eta = \eta(X_1, X_2) = \beta_0 + \beta_1(X_1 - \bar{X}_1) + \beta_2(X_2 - \bar{X}_2).$$

As is customary, we suppose the Y_i i.i.d. $N(\eta, \sigma^2)$. Here $\bar{X}_j$ has the usual meaning:

$$\bar{X}_j = \frac{1}{N} \sum_{i=1}^N X_{ji}, \quad j=1,2.$$

The following symbols are also defined for convenience

$$C_{jk} = \sum_{i=1}^N (X_{ji} - \bar{X}_j)(X_{ki} - \bar{X}_k), \quad j,k=1,2$$

and

$$D = (C_{11}C_{22} - C_{12}^2)^{-1}.$$

We shall use the usual least squares estimators b_0 , b_1 and b_2 for the β 's and $S^2 = \frac{1}{N-3} \sum_{i=1}^N (Y_i - \hat{\eta})^2$ for σ^2 where the caret has the usual least squares meaning.

We are now given a single value Y^0 for the dependent variable and asked to find its corresponding (X_1^0, X_2^0) and to bracket this set of (X_1, X_2) values by means of a confidence region. From the geometry in Fig. 1, which shows the situation in the positive octant, it is clear that the function of constant response $\hat{\eta} = Y^0$, on the regression plane, is the straight line A defined by

$$\begin{aligned} \text{A: } \hat{\eta} &= Y^0 \\ \hat{\eta} &= b_0 + b_1(X_1 - \bar{X}_1) + b_2(X_2 - \bar{X}_2). \end{aligned}$$

The line B is the vertical projection of the line A on the (X_1, X_2) plane and this is described by the equations

$$\begin{aligned} \text{B: } \hat{\eta} &= 0 \\ \hat{\eta} &= (b_0 - Y^0) + b_1(X_1 - \bar{X}_1) + b_2(X_2 - \bar{X}_2). \end{aligned}$$

Any point (X_1^0, X_2^0) on the line B will give a value $\hat{\eta} = Y^0$ when substituted into the regression equation (these are the points referred to previously as "corresponding to Y^0 ") and it is this set of points that has to be bracketed with a confidence region. In the case of the simple regression function the X^0 "corresponding" to a given value of Y^0 could be found explicitly. This cannot be done in the present case; however, this does not matter since the approach which will be used now is simply to make use of the fact, implicit in our model, that

$$Y^0 - \hat{\eta}_0,$$

where $\hat{\eta}_0 = \hat{\eta}(X_1^0, X_2^0) = b_0 + b_1(X_1^0 - \bar{X}_1) + b_2(X_2^0 - \bar{X}_2)$.

is distributed normally with mean value zero and variance

$$\sigma^2 f(X_1^0, X_2^0), \text{ where}$$

$$(2.1) \quad f(X_1^0, X_2^0) = \left(1 + \frac{1}{N}\right) + D[C_{22}(X_1^0 - \bar{X}_1)^2 - 2C_{12}(X_1^0 - \bar{X}_1)(X_2^0 - \bar{X}_2) + C_{11}(X_2^0 - \bar{X}_2)^2].$$

The random variable

$$\frac{Y^0 - \hat{\eta}_0}{S\sqrt{f(X_1^0, X_2^0)}}$$

will then be distributed as Student's t with $(N-3)$ degrees of freedom. The following probability statement is then valid:

$$(2.2) \quad P\{|Y^0 - \hat{\eta}_0| \leq t_{N-3}(\frac{1}{2}\alpha) \cdot S\sqrt{f(X_1^0, X_2^0)}\} = 1 - \alpha,$$

where $t_{N-3}(\frac{1}{2}\alpha)$ is the upper $\frac{1}{2}\alpha$ point of the t -distribution with $(N-3)$ d.f.

2.2 Analytic study of the confidence region

It is now clear that all points X_1, X_2 which satisfy the inequality

$$(Y^0 - \hat{\eta}(X_1, X_2))^2 \leq S^2 F_{1, N-3}(\alpha) f(X_1, X_2)$$

will form an $(1-\alpha)$ confidence region for the previously mentioned set of points (X_1^0, X_2^0) , where $F_{1, N-3}(\alpha)$ is the upper α point of the F -distribution with 1 and $(N-3)$ degrees of freedom. From now on we shall abbreviate this simply to F . This region has probability $(1-\alpha)$ for including the unknown (X_1^0, X_2^0) . Examining this region in greater detail we see that its boundary is described by the

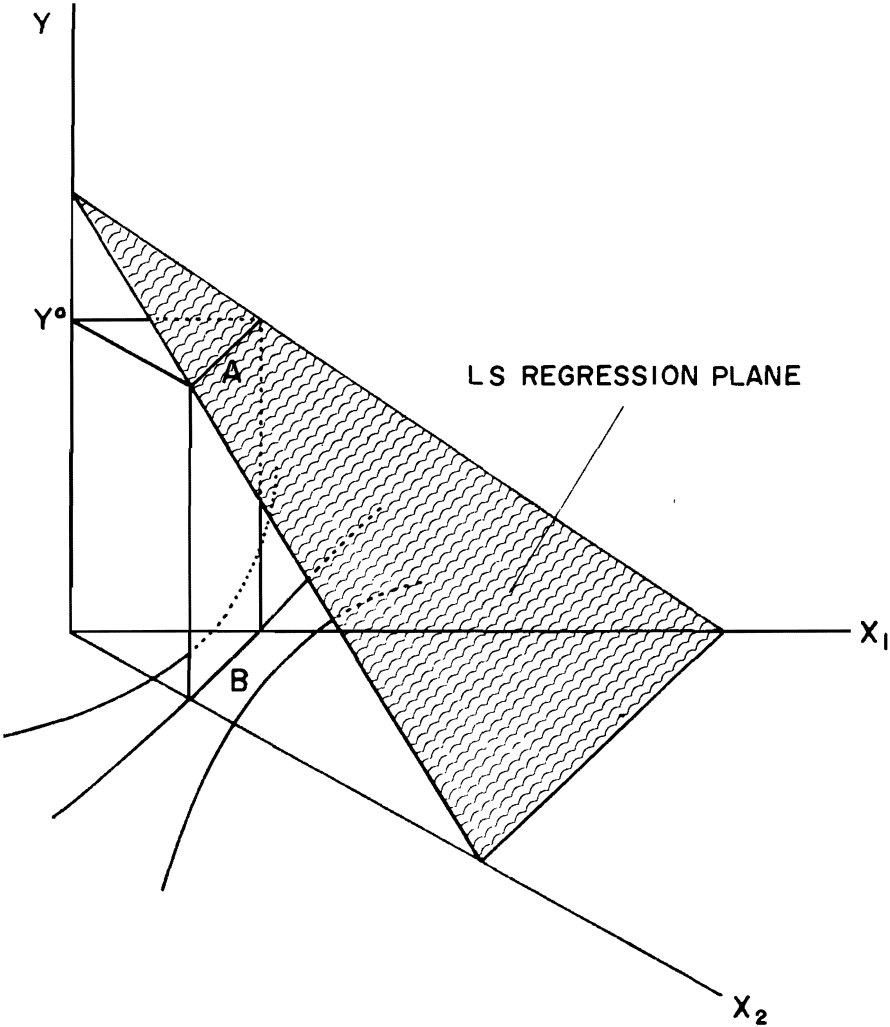


FIG. 1

quadratic equation $G(X,Y) = 0$ where $G(X,Y) \equiv aX^2 + 2hXY + bY^2 + 2gX + 2fY + c$ and X and Y are new Cartesian co-ordinates indicating that we have shifted the origin of our co-ordinate system to the centre of our experimental layout, i.e.

$$X = X_1 - \bar{X}_1 \quad \text{and} \quad Y = X_2 - \bar{X}_2.$$

The coefficients of the function G are as follows:

$$\begin{aligned} a &= b_1^2 - S^2_{FDC22} \\ b &= b_2^2 - S^2_{FDC11} \\ (2.3) \quad c &= (Y^0 - \bar{Y})^2 - S^2_{F(1+\frac{1}{N})} \\ f &= b_2(b_0 - Y^0) \\ g &= b_1(\bar{b}_0 - Y^0) \\ h &= b_1b_2 + S^2_{FDC12}. \end{aligned}$$

2.3 Geometric considerations

To determine the geometric form of the boundary it is only necessary to consider the value of the discriminant of the quadratic function $G(X,Y)$, namely $\Delta_2 = ab - h^2$. The coefficients of G are random variables and hence the probability that the curve is degenerate, is zero. The curve will in general be an ellipse or hyperbola (cf. Van Rooy (1946), p. 251) according to whether $\Delta_2 > 0$ or $\Delta_2 < 0$. The probability that $\Delta_2 = 0$, giving rise to a parabola, is zero.

Now it can be deduced that

$$\Delta_2 = 2S^4_{FD}(\frac{1}{2}F - T),$$

where T is a r.v. with an F -distribution with 2 and $N-3$ d.f., which is of the form $\Delta_2 = A - BT$, where A and B are positive constants. It follows that the confidence region being sought will be bounded by an ellipse if

$$T < A/B = \frac{1}{2}F$$

and otherwise by a hyperbola.

2.4 The plausibility of an elliptic confidence region

We are constructing a confidence region for the line B referred to in Section 2.1. Accordingly it does not seem intuitively reasonable that we should obtain a closed curve as the boundary of the confidence set of points. To indicate under what conditions such a closed curve may arise we re-examine the discriminant function Δ_2 . We have already stated that Δ_2 may be expressed in the form

$$\Delta_2 = -2S^4_{FD}(T - \frac{1}{2}F_{1,N-3}(\alpha))$$

and T may, for the present purpose, be written as

$$T = (C_{11}b_1^2 + 2C_{12}b_1b_2 + C_{22}b_2^2)/2S^2.$$

It is known (cf. Brownlee (1960), p. 476) that T is a test statistic (with null-distribution $F_{2,N-3}$) for the joint hypothesis $\beta_1 = \beta_2 = 0$. Now, if we test $\beta_1 = \beta_2 = 0$ by means of T at an α level of significance, the critical value will be $F_{2,N-3}(\alpha)$. It is a straightforward procedure to show from first principles that $2F_{2,N-3}(\alpha) > F_{1,N-3}(\alpha)$ and hence, if it turns out that $\Delta_2 > 0$ (indicating an elliptic confidence region for (X_1^0, X_2^0)), the test T cannot reject $\beta_1 = \beta_2 = 0$ at the α level of significance. This

means, except perhaps for a long-run fraction α of cases, that we are not dealing with true regression on 2 concomitant variables, in which case the present theory becomes irrelevant.

For a non-trivial linear regression on X_1 and X_2 we have established that the boundary of the confidence set we are seeking is described by the hyperbolic equation $G(X,Y)=0$, in the transformed variables X and Y . The set of points actually consists of all those satisfying $G(X,Y) \leq 0$. The question now arises whether this set of points is inside or outside the hyperbola. It may then be shown that we will actually be dealing with the set of points inside the hyperbola as is also intuitively obvious.

3. The use of regression in reverse for the mean value of k Y-observations

If we are told that $Y_1^0, \dots, Y_k^0$ are independent replicates of the experiment at the same (unknown, constant) level (X_1^0, X_2^0) of the independent variables and we are required to repeat what in the previous paragraph we have done for Y_1^0 , but this time for

$$\bar{Y}^0 = \frac{1}{k} \sum_{i=1}^k Y_i^0,$$

then the results are exactly the same except that in formulae (2.1) and (2.3) the term $(1 + \frac{1}{N})$ should be replaced by $(\frac{1}{k} + \frac{1}{N})$. This change should be effected in the rest of the formulae where $f(X_1^0, X_2^0)$ and C are used, but this leaves all results in section 2 basically unaltered.

4. Multiple discrimination with two independent variables

This time we have k observations $Y_1^0, \dots, Y_k^0$ which are independent of each other and for which we have to find k sets of corresponding values $(X_1^{0i}, X_2^{0i}), i=1, \dots, k$ and a joint confidence statement on these.

Making use of the Bonferroni inequality (cf. Miller (1966), p. 117) the probability statement (2.2) will change into

$$(4.1) \quad P\{|Y_i^0 - \bar{Y}^0(X_1^{0i}, X_2^{0i})| \leq t_{N-3}(\alpha/2k) \cdot S\sqrt{f(X_1^{0i}, X_2^{0i})}, \\ i=1, \dots, k\} \geq 1 - \alpha.$$

The overall effect of this on all ensuing formulae is that $F_{1, N-3}(\alpha)$ of section 2 must now be replaced by $F_{1, N-3}(\alpha/k)$.

If the Scheffé multiple comparison techniques are used in the above problem we get the same result except for a different critical constant. As may be seen from Miller (1966), p. 120 the probability statement (4.1) remains valid when $t_{N-3}(\alpha/2k)$ is replaced by the Scheffé constant $(k F_{k, N-3}(\alpha))^{\frac{1}{2}}$.

In those formulae to be derived from the inequalities inside the curly brackets of the probability statement the critical constant F of section 2 is now replaced by $k F_{k, N-3}(\alpha)$ for the Scheffé confidence regions. Nothing else is changed.

5. The multiple linear regression model

We now consider the usual model in multiple linear regression, namely

$$Y_j = \beta_0 + \beta_1 X_{1j} + \dots + \beta_p X_{pj} + e_j, \quad j=1, \dots, N.$$

The random variables $\{e_j\}$ are considered to have i.i.d. normal distributions with $E(e_j)=0$ and $\text{Var}(e_j)=\sigma^2$. The least squares estimate of β_i is denoted by b_i . The j -th observed value of the independent variable X_i is denoted by X_{ij} and X is the design matrix of the experiment, namely

$$X' = \begin{pmatrix} 1 & X_{11} & X_{21} & \cdots & X_{p1} \\ 1 & X_{12} & X_{22} & \cdots & X_{p2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1N} & X_{2N} & \cdots & X_{pN} \end{pmatrix}$$

and this is taken to be a matrix of full rank.

We are given a set of different values $Y_1^0, \dots, Y_k^0$ of the response variable and asked to find those sets of $(X_1^{0i}, \dots, X_p^{0i})$, $i=1, \dots, k$, corresponding to the Y_i^0 and to bracket these simultaneously in confidence intervals.

This time we consider the random variable

$$Y_i^0 - b_0 - b_1 X_1^{0i} - \dots - b_p X_p^{0i}$$

and observe that it is normally distributed with expected value zero and variance $\sigma^2[1 + X_0^{0i} C X_0^{0i}]$, (cf. Draper and Smith (1966), p. 121) where C is the inverse of the matrix $X'X$ and where X_0^{0i} is the vector variable corresponding to Y_i^0 . From the fact that a Student t -variable may now be constructed as in the previous sections for the case $p=2$, it is clear that the sets of points X_0^{0i} , $i=1, \dots, k$, defined by the Bonferroni regions,

$$G(X_1^{0i}, \dots, X_p^{0i}) \leq 0,$$

where $G = (Y_i^0 - X_0^{0i} b)^2 - S^2(1 + X_0^{0i} C X_0^{0i}) F_{1, N-p-1}(\alpha/k)$

will have simultaneous confidence coefficient $\geq 1 - \alpha$ and equality holds if $k=1$. In the case of Scheffé regions the critical constant again changes to $k F_{k, N-p-1}(\alpha)$.

6. Numerical Example

The theory considered in this paper originated from discussions with Mr. P.H. Müller of the Timber Research Unit of the South African Council for Scientific and Industrial Research, in connection with his practical problem, which is treated in this section. The data that are here subjected to analysis were gathered by the TRU. Specific gravity determinations (Y) were made on random samples of *Pinus patula* wood having a certain number of annual rings per inch of wood (X_1) and also at various levels of latewood (X_2) occurring in the tree from which a particular sample was taken. A total number $N=248$ of triplets of observation (Y, X_1, X_2) were recorded. The sample range of X_1 was between 3 and 6 and of X_2 between 5% and 40%. The specific gravity values ranged between 0.306 and 0.659.

It was required to determine those values of X_1 and X_2 which could reasonably have been associated with a value of $Y=0.39$. It was observed that the logarithm of the variable Y was approximately normally distributed and that approximate linear relationships existed between $Y'=\log Y$ and X_1 and $Y'=\log Y$ and X_2 . A linear regression was accordingly calculated between $Y'=\log Y$ and X_1 and X_2 and the standard calculations of linear regression resulted in the following set of values

$$\begin{aligned}
 N &= 248 \\
 \bar{Y}' &= -0.3879 \\
 \bar{X}_1 &= 2.6653 & \bar{X}_2 &= 15.7500 \\
 C_{11} &= 268.7218 & C_{12} &= 1329.7501 \\
 C_{22} &= 12720.5000 & D^{-1} &= 1650 \ 040.295
 \end{aligned}$$

Multiple Correlation Coefficient = 0.6038.

The regression equation has the following regression coefficients

$$b_0 = -0.3879, b_1 = 10^{-1} \times 0.2052, b_2 = 10^{-2} \times 0.2365$$

with variance about the regression plane

$$s^2 = 10^{-2} \times 0.2237.$$

A Y-value of 0.39 corresponds to a Y'-value of $Y_0' = -0.4089$ in the transformed space. Any point (X_1, X_2) on the line

$$X_1 = 3.4539 - 0.1152X_2$$

(in the (X_1, X_2) plane), when substituted into the regression plane, will yield a Y'-value of -0.4089. If we select $\alpha=0.05$, then the coefficients of the function $G(X, Y)$ as given in (2.3) are calculated as

$$\begin{aligned}
 a &= 10^{-3} \times 0.3545 \\
 b &= 10^{-5} \times 0.4185 \\
 c &= -10^{-2} \times 0.8240 \\
 f &= 10^{-4} \times 0.4982 \\
 g &= 10^{-3} \times 0.4323 \\
 h &= 10^{-4} \times 0.5551.
 \end{aligned}$$

Also, the discriminant of this quadratic function may be calculated as

$$\Delta_2 = -10^{-8} \times 0.1598 < 0$$

and hence the confidence region we are looking for is bounded by a hyperbola. The accompanying Fig. 2 sketches those parts of the hyperbola falling into the range of (X_1, X_2) observations. This chart may be used to make observations on X_1 and X_2 for any given sample of *P. patula* wood and if these values fall within the confidence region bounded by the hyperbola, then it may be concluded, without actually determining the S.G., that the wood sample has a specific gravity of 0.39. We could expect to be wrong on such statements in 5 cases out of 100 in the long run. Unfortunately, in the present example, the range of possible (X_1, X_2) -values is rather wide.

7. Acknowledgements

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REFERENCES

BROWNLEE, K.A. (1960). *Statistical theory and methodology in science and engineering*. John Wiley and Sons, New York.

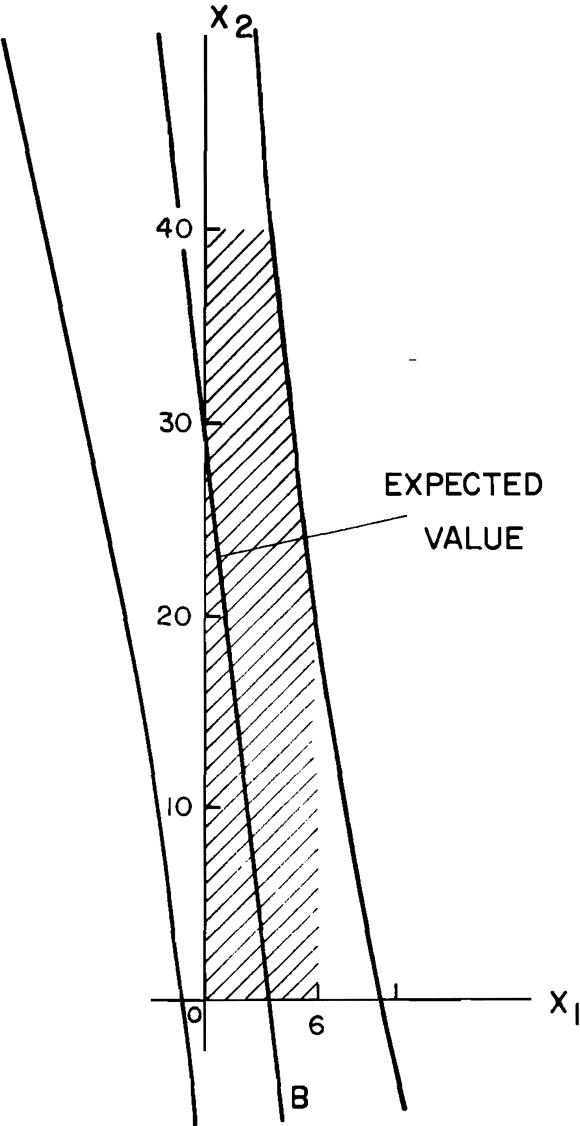


FIG. 2

- DRAPER, N.R. and SMITH, H. (1966). *Applied regression analysis*. John Wiley and Sons, New York.
- LAUBSCHER, N.F. (1962). Statistiese gevolgtrekkings by kwadratiese regressie. *Tydskr. Natuurwet.*, 2, 71-80.
- LIEBERMAN, G.F., MILLER, R.G. and HAMILTON, M.A. (1967). Unlimited simultaneous discrimination intervals in regression. *Biometrika*, 54, 133-146.
- MILLER, R.G. (1966). *Simultaneous statistical inference*. McGraw-Hill, New York.
- VAN ROOY, D.J. (1946). *Analitiese meetkunde van die plat vlak*. Stellenbosch, Pro Ecclesia-boekhandel.

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